

**TOPAZ MD4**

**Section: 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**

**1.1 Product identifier**

Product name : TOPAZ MD4  
Product code : 117576E  
Use of the Substance/Mixture : Foaming agent  
Substance type: : Mixture

**For professional users only.**

Product dilution information : No dilution information provided.

**1.2 Relevant identified uses of the substance or mixture and uses advised against**

Identified uses : Foam cleaner. Semi-Automatic without venting process  
Recommended restrictions on use : Reserved for industrial and professional use.

**1.3 Details of the supplier of the safety data sheet**

Company : Ecolab Temizleme Sistemleri Ltd. Şti  
Esentepe Mahallesi, Cevizli - Esentepe E-5 Yanyol Caddesi  
Vizyon Bulvarı No: 13, Kat 1 No: 65 Turkey TR 34870 KARTAL / İSTANBUL  
+90 (216) 458 69 00, Fax: +90 (216) 458 69 07

**1.4 Emergency telephone number**

Emergency telephone number : +32-(0)3-575-5555 Trans- European  
Poison Information Centre telephone number : 114 Ulusal Zehir Danışma Merkezi (UZEM)

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Version : 1.0

**Section: 2. HAZARDS IDENTIFICATION**

**2.1 Classification of the substance or mixture**

**Classification (T.R. SEA No 28848)**

|                                 |      |
|---------------------------------|------|
| Corrosive to metals, Category 1 | H290 |
| Skin corrosion, Sub-category 1A | H314 |
| Serious eye damage, Category 1  | H318 |

**TOPAZ MD4**

**2.2 Label elements**

**Labelling (T.R. SEA No 28848)**

Hazard pictograms :



Signal Word : Danger

Hazard Statements : H290 May be corrosive to metals.  
H314 Causes severe skin burns and eye damage.

Precautionary Statements : **Prevention:**  
P280 Wear protective gloves/ eye protection/ face protection.

**Response:**  
P303 + P361 + P353 IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower.  
P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.  
P310 Immediately call a POISON CENTER/doctor.

Hazardous components which must be listed on the label:  
sodium hydroxide

**2.3 Other hazards**

None known.

**Section: 3. COMPOSITION/INFORMATION ON INGREDIENTS**

**3.2 Mixtures**

**Hazardous components**

| Chemical Name                | CAS-No.<br>EC-No.       | Classification<br>(T.R. SEA No 28848)                                            | Concentration:<br>[%] |
|------------------------------|-------------------------|----------------------------------------------------------------------------------|-----------------------|
| sodium hydroxide             | 1310-73-2<br>215-185-5  | Skin corrosion Category 1A; H314<br>Corrosive to metals Category 1; H290         | >= 5 - < 10           |
| 2-(2-butoxyethoxy)ethanol    | 112-34-5<br>203-961-6   | Eye irritation Category 2; H319                                                  | >= 3 - < 5            |
| Sodiumcumenesulphonate       | 28348-53-0<br>248-983-7 | Eye irritation Category 2; H319                                                  | >= 3 - < 5            |
| Alcohols, C9-11, ethoxylated | 68439-46-3              | Serious eye damage Category 1; H318<br>Chronic aquatic toxicity Category 3; H412 | >= 1 - < 2.5          |
| Lauryl Alcohol               | 112-53-8<br>203-982-0   | Eye irritation Category 2; H319<br>Acute aquatic toxicity Category 1; H400       | >= 0.25 - < 0.5       |

**TOPAZ MD4**

|                        |                         |                                                                                                                                               |                     |
|------------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
|                        |                         | Chronic aquatic toxicity Category 2; H411                                                                                                     |                     |
| Lauryl Betaine         | 683-10-3<br>211-669-5   | Acute toxicity Category 3; H301<br>Acute toxicity Category 4; H312<br>Skin irritation Category 2; H315<br>Serious eye damage Category 1; H318 | $\geq 0.25 - < 0.5$ |
| Amphoteric surfactants | 97659-51-3<br>307-456-2 | Skin irritation Category 2; H315<br>Eye irritation Category 2; H319<br>Acute aquatic toxicity Category 1; H400                                | $\geq 0.25 - < 0.5$ |
| Myristyl Alcohol       | 112-72-1<br>204-000-3   | Eye irritation Category 2; H319<br>Chronic aquatic toxicity Category 1; H410                                                                  | $\geq 0.1 - < 0.25$ |

For the full text of the H-Statements mentioned in this Section, see Section 16.

**Section: 4. FIRST AID MEASURES**

**4.1 Description of first aid measures**

- In case of eye contact : Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention immediately.
- In case of skin contact : Wash off immediately with plenty of water for at least 15 minutes. Use a mild soap if available. Wash clothing before reuse. Thoroughly clean shoes before reuse. Get medical attention immediately.
- If swallowed : Rinse mouth with water. Do NOT induce vomiting. Never give anything by mouth to an unconscious person. If conscious, give 2 glasses of water. Get medical attention immediately.
- If inhaled : Remove to fresh air. Treat symptomatically. Get medical attention if symptoms occur.

**4.2 Most important symptoms and effects, both acute and delayed**

See Section 11 for more detailed information on health effects and symptoms.

**4.3 Indication of immediate medical attention and special treatment needed**

- Treatment : Treat symptomatically.

**Section: 5. FIREFIGHTING MEASURES**

**5.1 Extinguishing media**

- Suitable extinguishing media : Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
- Unsuitable extinguishing media : None known.

**5.2 Special hazards arising from the substance or mixture**

**TOPAZ MD4**

- Specific hazards during firefighting : Not flammable or combustible.
- Hazardous combustion products : Depending on combustion properties, decomposition products may include following materials:  
Carbon oxides  
Sulphur oxides  
metal oxides

**5.3 Advice for firefighters**

- Special protective equipment for firefighters : Use personal protective equipment.
- Further information : Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. In the event of fire and/or explosion do not breathe fumes.

**Section: 6. ACCIDENTAL RELEASE MEASURES**

**6.1 Personal precautions, protective equipment and emergency procedures**

- Advice for non-emergency personnel : Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Avoid inhalation, ingestion and contact with skin and eyes. When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed in sections 7 and 8.
- Advice for emergency responders : If specialised clothing is required to deal with the spillage, take note of any information in Section 8 on suitable and unsuitable materials.

**6.2 Environmental precautions**

- Environmental precautions : Do not allow contact with soil, surface or ground water.

**6.3 Methods and materials for containment and cleaning up**

- Methods for cleaning up : Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

**6.4 Reference to other sections**

- See Section 1 for emergency contact information.  
For personal protection see section 8.  
See Section 13 for additional waste treatment information.

**Section: 7. HANDLING AND STORAGE**

**TOPAZ MD4**

**7.1 Precautions for safe handling**

- Advice on safe handling : Do not ingest. Do not get in eyes, on skin, or on clothing. Use only with adequate ventilation. Wash hands thoroughly after handling. Do not breathe spray, vapour.
- Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Remove and wash contaminated clothing before re-use. Wash face, hands and any exposed skin thoroughly after handling. Provide suitable facilities for quick drenching or flushing of the eyes and body in case of contact or splash hazard.

**7.2 Conditions for safe storage, including any incompatibilities**

- Requirements for storage areas and containers : Do not store near acids. Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers.
- Absorb spillage to prevent material damage. Keep only in original packaging.
- Storage temperature : 0 °C to 40 °C
- Packaging material : Suitable material: Plastic material  
Unsuitable material: Aluminium, Mild steel

**7.3 Specific end uses**

- Specific use(s) : Foam cleaner. Semi-Automatic without venting process

**Section: 8. EXPOSURE CONTROLS/PERSONAL PROTECTION**

**8.1 Control parameters**

**Occupational Exposure Limits**

| Components                | CAS-No.  | Value type (Form of exposure)                                                               | Control parameters                | Basis  |
|---------------------------|----------|---------------------------------------------------------------------------------------------|-----------------------------------|--------|
| 2-(2-butoxyethoxy)ethanol | 112-34-5 | STEL 15 min                                                                                 | 15 ppm<br>101.2 mg/m <sup>3</sup> | TR OEL |
| Further information       |          | EU - I / C (1991 / 322 / EC) list of indicative limit values (updated with 2006 / 15 / EC ) |                                   |        |
|                           |          | TWA (8 Hour)                                                                                | 10 ppm<br>67.5 mg/m <sup>3</sup>  | TR OEL |
| Further information       |          | EU - I / C (1991 / 322 / EC) list of indicative limit values (updated with 2006 / 15 / EC ) |                                   |        |

**DNEL**

|                  |   |                                                                                                                                    |
|------------------|---|------------------------------------------------------------------------------------------------------------------------------------|
| sodium hydroxide | : | End Use: Workers<br>Exposure routes: Inhalation<br>Potential health effects: Long-term local effects<br>Value: 1 mg/m <sup>3</sup> |
|                  |   | End Use: Consumers<br>Exposure routes: Inhalation<br>Potential health effects: Long-term local effects                             |

**TOPAZ MD4**

|                           |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           |   | Value: 1 mg/m3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 2-(2-butoxyethoxy)ethanol | : | <p>End Use: Workers<br/>Exposure routes: Inhalation<br/>Potential health effects: Short-term - local<br/>Value: 101.2 mg/m3</p> <p>End Use: Workers<br/>Exposure routes: Dermal<br/>Potential health effects: Long-term systemic effects<br/>Value: 20 mg/kg</p> <p>End Use: Workers<br/>Exposure routes: Inhalation<br/>Potential health effects: Long-term systemic effects<br/>Value: 67.5 mg/m3</p> <p>End Use: Workers<br/>Exposure routes: Inhalation<br/>Potential health effects: Short-term - local<br/>Value: 67.5 mg/m3</p> |

**PNEC**

|                           |   |                                                                                                                                                                                                                                                                                        |
|---------------------------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2-(2-butoxyethoxy)ethanol | : | <p>Fresh water<br/>Value: 1 mg/l</p> <p>Marine water<br/>Value: 0.1 mg/l</p> <p>Intermittent use/release<br/>Value: 3.9 mg/l</p> <p>Sewage treatment plant<br/>Value: 200 mg/l</p> <p>Sediment<br/>Value: 4 mg/kg</p> <p>Soil<br/>Value: 0.4 mg/kg</p> <p>Oral<br/>Value: 56 mg/kg</p> |
|---------------------------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**8.2 Exposure controls**

**Appropriate engineering controls**

Engineering measures : Effective exhaust ventilation system. Maintain air concentrations below occupational exposure standards.

**TOPAZ MD4**

**Individual protection measures**

- Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Remove and wash contaminated clothing before re-use. Wash face, hands and any exposed skin thoroughly after handling. Provide suitable facilities for quick drenching or flushing of the eyes and body in case of contact or splash hazard.
- Eye/face protection (EN 166) : Safety goggles  
Face-shield
- Hand protection (EN 374) : Recommended preventive skin protection  
Gloves  
Nitrile rubber  
butyl-rubber  
Breakthrough time: 1 – 4 hours  
Minimum thickness for butyl-rubber 0.7 mm for nitrile rubber 0.4 mm or equivalent (please refer to the gloves manufacturer/distributor for advise).  
Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.
- Skin and body protection (EN 14605) : Personal protective equipment comprising: suitable protective gloves, safety goggles and protective clothing including appropriate safety shoes
- Respiratory protection (EN 143, 14387) : None required if airborne concentrations are maintained below the exposure limit listed in Exposure Limit Information. Use certified respiratory protection equipment meeting EU requirements(89/656/EEC, (EU) 2016/425), or equivalent, when respiratory risks cannot be avoided or sufficiently limited by technical means of collective protection or by measures, methods or procedures of work organization.

**Environmental exposure controls**

- General advice : Consider the provision of containment around storage vessels.

**Section: 9. PHYSICAL AND CHEMICAL PROPERTIES**

**9.1 Information on basic physical and chemical properties**

- Appearance : liquid
- Colour : clear, yellow
- Odour : amine-like
- pH : 12.5 - 13.5, 100 %
- Flash point : Not applicable., Does not sustain combustion.
- Odour Threshold : Not applicable and/or not determined for the mixture
- Melting point/freezing point : Not applicable and/or not determined for the mixture
- Initial boiling point and : > 100 °C

**TOPAZ MD4**

boiling range

|                                        |                                                            |
|----------------------------------------|------------------------------------------------------------|
| Evaporation rate                       | : Not applicable and/or not determined for the mixture     |
| Flammability (solid, gas)              | : Not applicable and/or not determined for the mixture     |
| Upper explosion limit                  | : Not applicable and/or not determined for the mixture     |
| Lower explosion limit                  | : Not applicable and/or not determined for the mixture     |
| Vapour pressure                        | : Not applicable and/or not determined for the mixture     |
| Relative vapour density                | : Not applicable and/or not determined for the mixture     |
| Relative density                       | : 1.14 - 1.15                                              |
| Water solubility                       | : soluble                                                  |
| Solubility in other solvents           | : Not applicable and/or not determined for the mixture     |
| Partition coefficient: n-octanol/water | : Not applicable and/or not determined for the mixture     |
| Auto-ignition temperature              | : Not applicable and/or not determined for the mixture     |
| Thermal decomposition                  | : Not applicable and/or not determined for the mixture     |
| Viscosity, kinematic                   | : Not applicable and/or not determined for the mixture     |
| Explosive properties                   | : Not applicable and/or not determined for the mixture     |
| Oxidizing properties                   | : The substance or mixture is not classified as oxidizing. |

**9.2 Other information**

Not applicable and/or not determined for the mixture

**Section: 10. STABILITY AND REACTIVITY**

**10.1 Reactivity**

No dangerous reaction known under conditions of normal use.

**10.2 Chemical stability**

Stable under normal conditions.

**10.3 Possibility of hazardous reactions**

No dangerous reaction known under conditions of normal use.

**10.4 Conditions to avoid**

None known.

**10.5 Incompatible materials**

Acids

Aluminium  
Mild steel

**TOPAZ MD4**

**10.6 Hazardous decomposition products**

Depending on combustion properties, decomposition products may include following materials:

Carbon oxides  
Sulphur oxides  
metal oxides

**Section: 11. TOXICOLOGICAL INFORMATION**

**11.1 Information on toxicological effects**

Information on likely routes of exposure : Inhalation, Eye contact, Skin contact

**Product**

|                                   |                                                |
|-----------------------------------|------------------------------------------------|
| Acute oral toxicity               | : Acute toxicity estimate : > 2,000 mg/kg      |
| Acute inhalation toxicity         | : There is no data available for this product. |
| Acute dermal toxicity             | : There is no data available for this product. |
| Skin corrosion/irritation         | : There is no data available for this product. |
| Serious eye damage/eye irritation | : There is no data available for this product. |
| Respiratory or skin sensitization | : There is no data available for this product. |
| Carcinogenicity                   | : There is no data available for this product. |
| Reproductive effects              | : There is no data available for this product. |
| Germ cell mutagenicity            | : There is no data available for this product. |
| Teratogenicity                    | : There is no data available for this product. |
| STOT - single exposure            | : There is no data available for this product. |
| STOT - repeated exposure          | : There is no data available for this product. |
| Aspiration toxicity               | : There is no data available for this product. |

**Components**

|                     |                                                       |
|---------------------|-------------------------------------------------------|
| Acute oral toxicity | : 2-(2-butoxyethoxy)ethanol<br>LD50 rat: 3,306 mg/kg  |
|                     | Sodiumcumenesulphonate<br>LD50 rat: > 7,000 mg/kg     |
|                     | Alcohols, C9-11, ethoxylated<br>LD50 rat: 4,044 mg/kg |

**TOPAZ MD4**

Lauryl Alcohol  
LD50 rat: > 5,000 mg/kg

Lauryl Betaine  
LD50 rat: 71 mg/kg

Amphoteric surfactants  
LD50 rat: > 5,000 mg/kg

**Components**

Acute inhalation toxicity : Sodiumcumenesulphonate  
4 h LC50 rat: > 770 mg/l  
Test atmosphere: dust/mist

Lauryl Alcohol  
4 h LC50 rat: 35.5 mg/l  
Test atmosphere: dust/mist

**Components**

Acute dermal toxicity : 2-(2-butoxyethoxy)ethanol  
LD50 rabbit: 2,764 mg/kg

Sodiumcumenesulphonate  
LD50 rabbit: > 2,000 mg/kg

Alcohols, C9-11, ethoxylated  
LD50 rabbit: 2,108 mg/kg  
Test substance: Information given is based on data obtained from similar substances.

Lauryl Alcohol  
LD50 rabbit: 8,000 mg/kg

Lauryl Betaine  
LD50 rat: 1,300 mg/kg

Myristyl Alcohol  
LD50 rabbit: 8,000 mg/kg

**Potential Health Effects**

Eyes : Causes serious eye damage.

Skin : Causes severe skin burns.

Ingestion : Causes digestive tract burns.

Inhalation : May cause nose, throat, and lung irritation.

Chronic Exposure : Health injuries are not known or expected under normal use.

**Experience with human exposure**

**TOPAZ MD4**

Eye contact : Redness, Pain, Corrosion

Skin contact : Redness, Pain, Corrosion

Ingestion : Corrosion, Abdominal pain

Inhalation : Respiratory irritation, Cough

**Section: 12. ECOLOGICAL INFORMATION**

**12.1 Ecotoxicity**

Environmental Effects : This product has no known ecotoxicological effects.

**Product**

Toxicity to fish : no data available

Toxicity to daphnia and other aquatic invertebrates : no data available

Toxicity to algae : no data available

**Components**

Toxicity to fish : 2-(2-butoxyethoxy)ethanol  
96 h LC50 Fish: 1,300 mg/l

Sodiumcumenesulphonate  
96 h LC50 Fish: > 450 mg/l

Alcohols, C9-11, ethoxylated  
96 h LC50 Oncorhynchus mykiss (rainbow trout): 5 mg/l

Myristyl Alcohol  
96 h LC50 Oncorhynchus mykiss (rainbow trout): > 1 mg/l

**Components**

Toxicity to daphnia and other aquatic invertebrates : sodium hydroxide  
48 h EC50: 40 mg/l

Alcohols, C9-11, ethoxylated  
48 h EC50 Daphnia magna (Water flea): 2.5 mg/l

Lauryl Alcohol  
48 h EC50 Daphnia: 0.77 mg/l

Myristyl Alcohol  
48 h EC50 Daphnia magna (Water flea): 3.2 mg/l

**Components**

Toxicity to algae : Alcohols, C9-11, ethoxylated  
96 h EC50 Pseudokirchneriella subcapitata (algae): 1.4 mg/l

Myristyl Alcohol  
96 h EC50 Desmodesmus subspicatus (green algae): > 10 mg/l

**TOPAZ MD4**

**12.2 Persistence and degradability**

**Product**

Biodegradability : The surfactants contained in the product are biodegradable according to the requirements of the detergent regulation 648/2004/EC

**Components**

Biodegradability : sodium hydroxide  
Result: Not applicable - inorganic

2-(2-butoxyethoxy)ethanol  
Result: Readily biodegradable.

Sodiumcumenesulphonate  
Result: Readily biodegradable.

Alcohols, C9-11, ethoxylated  
Result: Readily biodegradable.

Lauryl Alcohol  
Result: Readily biodegradable.

Myristyl Alcohol  
Result: Readily biodegradable.

**12.3 Bioaccumulative potential**

no data available

**12.4 Mobility in soil**

no data available

**12.5 Results of PBT and vPvB assessment**

**Product**

Assessment : This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

**12.6 Other adverse effects**

no data available

**Section: 13. DISPOSAL CONSIDERATIONS**

Dispose of in accordance with the European Directives on waste and hazardous waste. Waste codes should be assigned by the user, preferably in discussion with the waste disposal authorities.

**13.1 Waste treatment methods**

**TOPAZ MD4**

- Product : Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.
- Contaminated packaging : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.
- Guidance for Waste Code selection : Inorganic wastes containing dangerous substances. If this product is used in any further processes, the final user must redefine and assign the most appropriate European Waste Catalogue Code. It is the responsibility of the waste generator to determine the toxicity and physical properties of the material generated to determine the proper waste identification and disposal methods in compliance with applicable European (EU Directive 2008/98/EC) and local regulations.

**Section: 14. TRANSPORT INFORMATION**

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

**Land transport (ADR/ADN/RID)**

- 14.1 UN number : 1824
- 14.2 UN proper shipping name : SODIUM HYDROXIDE SOLUTION
- 14.3 Transport hazard class(es) : 8
- 14.4 Packing group : II
- 14.5 Environmental hazards : No
- 14.6 Special precautions for user : None

**Air transport (IATA)**

- 14.1 UN number : 1824
- 14.2 UN proper shipping name : Sodium hydroxide solution
- 14.3 Transport hazard class(es) : 8
- 14.4 Packing group : II
- 14.5 Environmental hazards : No
- 14.6 Special precautions for user : None

**Sea transport (IMDG/IMO)**

- 14.1 UN number : 1824
- 14.2 UN proper shipping name : SODIUM HYDROXIDE SOLUTION
- 14.3 Transport hazard class(es) : 8
- 14.4 Packing group : II

**TOPAZ MD4**

14.5 Environmental hazards : No  
14.6 Special precautions for user : None  
14.7 Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code : Not applicable.

**Section: 15. REGULATORY INFORMATION**

**15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture**

according to Detergents Regulation EC 648/2004 : less than 5 %: Phosphonates, Anionic surfactants, Cationic surfactants, Amphoteric surfactants, Non-ionic surfactants, Polycarboxylates

**National Regulations**

**Take note of Dir 94/33/EC on the protection of young people at work.**

Other regulations : According to 11 December 2013, Numbered 28848 (Bis), "Ministry of Environment and Forestry"; Regulation on Classification, Labelling and Packaging of Substances and Mixtures.  
According to 13 Dec 2014, Numbered 29204, "Ministry of Environment and Urbanization"; Regulation on Safety Data Sheets regarding Dangerous Substances and Mixtures.

**15.2 Chemical Safety Assessment**

No Chemical Safety Assessment has been carried out on the product.

**Section: 16. OTHER INFORMATION**

**Procedure used to derive the classification according to REGULATION (EC) No 1272/2008 and Regulation T.R. SEA No 28848**

| Classification              | Justification      |
|-----------------------------|--------------------|
| Corrosive to metals 1, H290 | Calculation method |
| Skin corrosion 1A, H314     | Calculation method |
| Serious eye damage 1, H318  | Calculation method |

**Full text of H-Statements**

H290 May be corrosive to metals.  
H301 Toxic if swallowed.  
H312 Harmful in contact with skin.  
H314 Causes severe skin burns and eye damage.  
H315 Causes skin irritation.  
H318 Causes serious eye damage.  
H319 Causes serious eye irritation.  
H400 Very toxic to aquatic life.  
H410 Very toxic to aquatic life with long lasting effects.  
H411 Toxic to aquatic life with long lasting effects.  
H412 Harmful to aquatic life with long lasting effects.

**TOPAZ MD4**

**Full text of other abbreviations**

ADN – European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways; ADR – European Agreement concerning the International Carriage of Dangerous Goods by Road; AICS – Australian Inventory of Chemical Substances; ASTM – American Society for the Testing of Materials; bw – Body weight; CLP – Classification Labelling Packaging Regulation; Regulation (EC) No 1272/2008; CMR – Carcinogen, Mutagen or Reproductive Toxicant; DIN – Standard of the German Institute for Standardisation; DSL – Domestic Substances List (Canada); ECHA – European Chemicals Agency; EC-Number – European Community number; ECx – Concentration associated with x% response; ELx – Loading rate associated with x% response; EmS – Emergency Schedule; ENCS – Existing and New Chemical Substances (Japan); ErCx – Concentration associated with x% growth rate response; GHS – Globally Harmonized System; GLP – Good Laboratory Practice; IARC – International Agency for Research on Cancer; IATA – International Air Transport Association; IBC – International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 – Half maximal inhibitory concentration; ICAO – International Civil Aviation Organization; IECSC – Inventory of Existing Chemical Substances in China; IMDG – International Maritime Dangerous Goods; IMO – International Maritime Organization; ISHL – Industrial Safety and Health Law (Japan); ISO – International Organisation for Standardization; KECI – Korea Existing Chemicals Inventory; LC50 – Lethal Concentration to 50 % of a test population; LD50 – Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL – International Convention for the Prevention of Pollution from Ships; n.o.s. – Not Otherwise Specified; NO(A)EC – No Observed (Adverse) Effect Concentration; NO(A)EL – No Observed (Adverse) Effect Level; NOELR – No Observable Effect Loading Rate; NZIoC – New Zealand Inventory of Chemicals; OECD – Organization for Economic Co-operation and Development; OPPTS – Office of Chemical Safety and Pollution Prevention; PBT – Persistent, Bioaccumulative and Toxic substance; PICCS – Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR – (Quantitative) Structure Activity Relationship; REACH – Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RID – Regulations concerning the International Carriage of Dangerous Goods by Rail; SADT – Self-Accelerating Decomposition Temperature; SDS – Safety Data Sheet; TCSI – Taiwan Chemical Substance Inventory; TRGS – Technical Rule for Hazardous Substances; TSCA – Toxic Substances Control Act (United States); UN – United Nations; vPvB – Very Persistent and Very Bioaccumulative

Prepared by : Name, Surname: Nuryıl Subaşı  
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Certificate date: 09.05.2018  
Certificate expiry date: 09.05.2021  
Contact: +902164586983

Numbers quoted in the MSDS are given in the format: 1,000,000 = 1 million and 1,000 = 1 thousand. 0.1 = 1 tenth and 0.001 = 1 thousandth

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the SDS.

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.